

DEPLOYMENT

1. Overview

The deployment files are a critical part of the ExoHabitAI project, as they enable the application to run in a production environment. These files ensure that the system is properly configured, all dependencies are installed, and the application can handle user requests efficiently.

The deployment setup allows the application to be hosted on a cloud platform, making it accessible through a web browser without requiring any local installation. It ensures scalability, reliability, and smooth execution of the system.

2. app.py (Application File)

The app.py file is the core backend component of the application. It is responsible for initializing the Flask web server and managing all the functionalities of the system.

This file handles incoming requests from the frontend, processes user input data, and performs predictions using the trained machine learning model. It also ensures proper routing of requests through defined API endpoints such as prediction and data handling.

Additionally, the app.py file integrates the machine learning model into the web application, enabling real-time prediction of exoplanet habitability.

3. requirements.txt (Dependencies File)

The requirements.txt file contains a list of all Python libraries and dependencies required to run the application successfully.

Key libraries included are:

- Flask – for backend web development
- NumPy and Pandas – for data processing and manipulation
- Scikit-learn – for machine learning model implementation
- Gunicorn – for running the application in a production environment

This file plays an important role in deployment, as it allows the hosting platform to automatically install all necessary packages before running the application.

4. Procfile (Server Configuration File)

The Procfile is used to define the command required to run the application on the server. It specifies that the Flask application should be executed using Gunicorn, which is a production-level web server.

Using Gunicorn improves the performance and scalability of the application, allowing it to handle multiple user requests efficiently. It also ensures that the application runs continuously without interruptions.

5. runtime.txt (Runtime Configuration)

The runtime.txt file specifies the Python version required to run the application. This ensures compatibility between the development and deployment environments.

By maintaining a consistent runtime environment, this file helps prevent errors related to version mismatches and ensures stable performance of the application.

6. Model File (model.pkl)

The model.pkl file contains the trained machine learning model used for predicting the habitability of exoplanets.

This file is loaded by the backend during execution and is used to generate predictions based on user input. It plays a crucial role in the functioning of the application, as it directly impacts the accuracy and reliability of the results.

7. Deployment Process

The deployment process involves preparing all required files and uploading them to a cloud hosting platform such as Render.

The platform reads the requirements.txt file to install dependencies and uses the Procfile to execute the application using Gunicorn. The runtime environment is set using the runtime.txt file.

Once deployed, the application becomes accessible through a public URL, allowing users to interact with the system in real time. The deployment ensures that the application is scalable, reliable, and easily accessible.

8. Key Features of Deployment

The deployment setup offers several advantages:

- Enables access to the application from anywhere using a web browser
- Eliminates the need for local installation
- Ensures faster processing and real-time predictions
- Provides scalability to handle multiple users
- Simplifies maintenance and updates

9. Advantages of Deployment Files

The deployment files organize all necessary components required to run the application in a structured manner. They simplify the setup process and ensure that the application can be easily shared and hosted.

They also improve portability, allowing the application to run on different systems without modification. Additionally, they enhance the reliability and efficiency of the system.

10. Conclusion

The deployment files play a vital role in making the ExoHabitAI application functional in a real-world environment. They ensure proper configuration, smooth execution, and easy accessibility of the system.

By organizing all essential components and enabling cloud deployment, these files contribute significantly to the overall success and usability of the project.